

Angel Singh Shrestha

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Arlington, TX

EDUCATION

University of Texas at Arlington

B.S. Computer Science, GPA: 3.89

Arlington, TX

Expected May 2028

CERTIFICATIONS

AWS Certified AI Practitioner (Aug. 7, 2026) | AWS Certified Cloud Practitioner (Aug. 14, 2026) | Microsoft Certified: GitHub Actions (Aug. 28, 2026) | AWS Certified Solutions Architect (Expected Oct. 9, 2026)

EXPERIENCE

Student Technical Assistant | Network Operations

Feb. 2026 – Present

Office of Information Technology, University of Texas at Arlington

Arlington, TX

- Managed observability and incident-response workflows across hybrid Azure/on-prem infrastructure using Zabbix, Grafana, Splunk, and PagerDuty; maintained a **90%+ same-day resolution rate** for cross-functional escalations.
- Diagnosed Linux, application, and network infrastructure incidents through centralized log analysis and system telemetry; evaluated production changes through Change Advisory Board (CAB) processes.
- Engineered browser-based JavaScript automation for internal operational workflows, reducing repetitive manual processes for Tier 1 support teams.

Undergraduate Research Assistant

May 2025 – Jul. 2025

University of Texas at Arlington

Arlington, TX

- Developed a Python-based simulation framework via Jupyter to calculate node-level mesh impedance for 2D in-memory computing architectures, delivering computational models for Electronic Design Analysis (EDA) workflow analysis.

PROJECTS

PhilosoStream | GitHub | TypeScript, Next.js, FastAPI, MongoDB, AWS, Terraform, Docker

- Architected a full-stack note intelligence platform using Next.js/TypeScript, FastAPI, OAuth 2.0, MongoDB, Amazon S3, and Redis; designed secure REST APIs and caching strategies achieving **42 ms average API latency**.
- Built RAG and ML recommendation pipelines with vector embeddings, semantic retrieval, Apache Kafka, Airflow, and MLflow; served grounded LLM question answering, summarization, and personalized recommendations through dedicated inference services.
- Engineered asynchronous workflows with Amazon SNS/SQS/Lambda, Docker, Terraform, pytest, and GitHub Actions CI/CD; provisioned an auto-scaling Amazon ECS architecture designed for **5,000+ concurrent sessions** with retry and failure-recovery handling.

Cloud & Infrastructure Homelab | Linux, Docker, Proxmox, AWS, Azure, Nginx, Prometheus, Grafana

- Operated 10+ virtualized and containerized services across hybrid-cloud infrastructure spanning Proxmox, AWS, and Azure; implemented network segmentation, reverse proxying, secure remote access, and centralized observability with Prometheus, Grafana, Zabbix, and Splunk.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Go, TypeScript, JavaScript, Java, SQL, Bash

Backend & Frameworks: FastAPI, .NET/ASP.NET Core, Node.js, Next.js, REST APIs, gRPC, Apache Kafka, Redis, Microservices

AI/ML & Agentic Development: PyTorch, RAG, LLMs, Embeddings, Vector Search, Amazon Bedrock, OpenAI/Anthropic APIs, MCP, Claude Code

Cloud & DevOps: AWS (ECS, Lambda, S3, SQS, SNS, EC2), Azure, Docker, Kubernetes, Terraform, Linux, GitHub Actions

Databases & Observability: PostgreSQL, MySQL, MongoDB, DynamoDB, Prometheus, Grafana, Splunk, PagerDuty

Testing & Tools: Git, CMake, pytest, GoogleTest, Jest, Playwright, k6, Unit Testing, Integration Testing

RELEVANT COURSEWORK

Data Structures & Algorithms, Databases, Intro to Software Engineering, Programming Languages, Operating Systems, Object-Oriented Programming